DEEPAK R

Computer Science Engineering Student | Python Developer | AI Enthusiast | Systems Programmer

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PROFESSIONAL SUMMARY

Motivated Computer Science Engineering student with hands-on experience in Python development, Artificial Intelligence, and Computer Vision, combined with a strong foundation in Operating Systems, Computer Architecture, Networking, and Database Management Systems. Skilled in building real-world software solutions including facial recognition systems, CPU scheduling simulators, secure file management platforms, and machine learning forecasting applications. Seeking an internship to apply software development skills and contribute to intelligent, secure technology solutions.

TECHNICAL SKILLS

Languages	Python, C, C++, Java, SQL
AI & Computer Vision	OpenCV, LBPH Face Recognition, Haar Cascade Detection, Face Recognition, Machine Learning
Databases	MySQL, JDBC, Database Design, DBMS
OS & Architecture	CPU Scheduling, Operating Systems, Computer Architecture, Instruction Set Design
Networking	Wi-Fi 6, Cisco Packet Tracer, Network Optimization
Tools & Platforms	GitHub, VS Code, Qt 6, Power BI

PROJECTS

Smart Attendance Management System

Python, OpenCV, LBPH, Haar Cascade, Computer Vision

Built an AI-powered facial recognition attendance system that automates student identification and attendance logging.

Key Features: Face detection & recognition, liveness verification, student database integration, automated CSV reports

Impact: Reduced manual attendance effort and improved tracking accuracy through automated recognition

Wi-Fi 6 WLAN Architecture

Cisco Packet Tracer, Networking

Designed and optimized a modern Wi-Fi 6 wireless network architecture for enterprise-scale connectivity.

Key Features: OFDMA & MU-MIMO implementation, network topology design, performance optimization, Packet Tracer simulation

Impact: Demonstrated improved network efficiency and scalability for high-density wireless environments

Secure File Management System

Java, MySQL, AES Encryption, SHA-256

Developed a secure file management platform with encrypted storage and controlled access.

Key Features: Role-based access control (RBAC), AES encryption, SHA-256 integrity verification, user authentication

Impact: Strengthened data confidentiality and access control for sensitive file operations

Responsive CPU Task Scheduler

Python, Machine Learning

Built an intelligent CPU scheduling simulator to compare and analyze scheduling algorithm performance.

Key Features: FCFS, Round Robin & Priority scheduling, adaptive ML-based analysis, performance comparison dashboard

Impact: Improved understanding of resource allocation and scheduling efficiency trade-offs

Custom CPU Simulator

Python, Computer Architecture

Developed an interactive CPU simulator supporting custom instruction set creation and execution visualization. The simulator allows users to design instruction sets, simulate execution flow, and understand CPU architecture concepts through practical implementation.

Key Features: Custom instruction set design, CPU execution simulation, register and memory visualization, interactive learning environment

Impact: Improved understanding of processor architecture, instruction execution, and system-level programming concepts through hands-on simulation

Bike Demand Prediction

Python, Machine Learning, Power BI

Created a machine learning model to forecast bike-sharing demand using historical usage data.

Key Features: Predictive modeling, data preprocessing & feature engineering, Power BI visualization dashboard

Impact: Enabled data-driven demand forecasting to support better resource planning

EDUCATION

Bachelor of Engineering (B.E.) — Computer Science and Engineering

Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences (SIMATS), Chennai, India

Current Student • Expected Graduation: 2029

ACHIEVEMENTS

- Developed AI-powered attendance management system using computer vision
- Built CPU scheduling simulator for operating system concepts
- Designed secure file management system with encryption and RBAC
- Developed machine learning forecasting applications
- Designed Wi-Fi 6 network architecture using Cisco Packet Tracer
- Built multiple software engineering and system development projects
- Developed projects across Artificial Intelligence, Computer Vision, Operating Systems, Computer Architecture, Networking, and Database Systems